



Hippo–YAP Signaling Drives Epigenetic Vulnerability in the Diabetic Heart via the miR-22-3p–SIRT1 Axis: Linking Metabolic Stress to Chromatin Control

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Diabetic cardiomyopathy (DCM) is increasingly recognized as a distinct myocardial disease entity, characterized by early diastolic dysfunction, progressive interstitial fibrosis, cardiomyocyte hypertrophy, and eventual systolic failure, that occurs independently of epicardial coronary artery disease or chronic pressure overload. Epidemiological studies have shown that diabetes confers a two- to fourfold increased risk of heart failure even after adjustment for hypertension and ischemic heart disease, underscoring diabetes-specific myocardial vulnerability rather than simple acceleration of traditional cardiomyopathy pathways (1).

Despite intensive investigation, therapeutic progress in DCM has been limited. Tight glycemic control alone fails to prevent heart failure development, indicating that chronic hyperglycemia acts not merely as a metabolic insult but as a persistent signaling stress that reshapes cardiomyocyte transcriptional, epigenetic, and metabolic programs (2). This realization has shifted attention toward stress-responsive signaling networks capable of integrating metabolic cues with long-term remodeling responses, including the Hippo signaling pathway, which has emerged as a master regulator of tissue growth, regeneration, and fibrosis (3). The transcriptional coactivator Yes-associated protein (YAP) is the central effector of this pathway, shuttling between the cytoplasm and nucleus in response to mechanical stress, metabolic state, and cellular energy balance. When nuclear, YAP partners with TEAD transcription factors to induce gene programs governing cell survival, extracellular matrix remodeling, and inflammation (4).

In the cardiovascular system, YAP signaling exhibits striking context dependence. During development and neonatal life, YAP activation promotes cardiomyocyte proliferation and regenerative growth. In contrast, in the adult heart, sustained YAP activation has been linked to pathological hypertrophy, fibroblast activation, and adverse remodeling in pressure overload and ischemic injury models (5). Whether

YAP plays a similarly maladaptive role in metabolic cardiomyopathy has remained uncertain. Parallel to advances in Hippo biology, epigenetic regulators have gained prominence in DCM research. Sirtuin-1 (SIRT1), an NAD⁺-dependent deacetylase, is particularly notable. SIRT1 coordinates mitochondrial biogenesis, oxidative stress resistance, autophagy, and insulin sensitivity, processes consistently disrupted in diabetic hearts. Cardiac-specific SIRT1 deletion recapitulates key features of DCM, including hypertrophy and impaired relaxation, whereas pharmacological or genetic SIRT1 activation mitigates fibrosis and improves cardiac function (6).

MicroRNAs add another layer of regulation by fine-tuning stress-responsive gene networks (7). For instance, miR-22 has been repeatedly implicated in cardiac hypertrophy and metabolic regulation. Cardiac overexpression of miR-22 induces hypertrophy, whereas its inhibition improves stress tolerance (8). Of note, miR-22 directly targets SIRT1 in metabolic tissues, linking it to energy homeostasis (9).

In the current issue of *Diabetes*, Hou and associates (10) elegantly integrate these previously disconnected strands into a coherent pathogenic axis, demonstrating that hyperglycemia-induced YAP activation upregulates miR-22-3p, which in turn represses SIRT1, driving hypertrophy and fibrosis in DCM.

Hou and colleagues provide compelling evidence that YAP activation is not a bystander but a causal contributor to diabetic myocardial remodeling (10). Increased nuclear YAP accumulation in diabetic mouse hearts and glucolipotoxic cardiomyocytes mirrors observations in failing human hearts with metabolic disease (11).

Importantly, cardiac-restricted YAP silencing or genetic ablation improves both systolic and diastolic function, reduces cardiomyocyte cross-sectional area, and attenuates collagen deposition (10). These findings align with prior demonstrations that YAP promotes fibroblast activation

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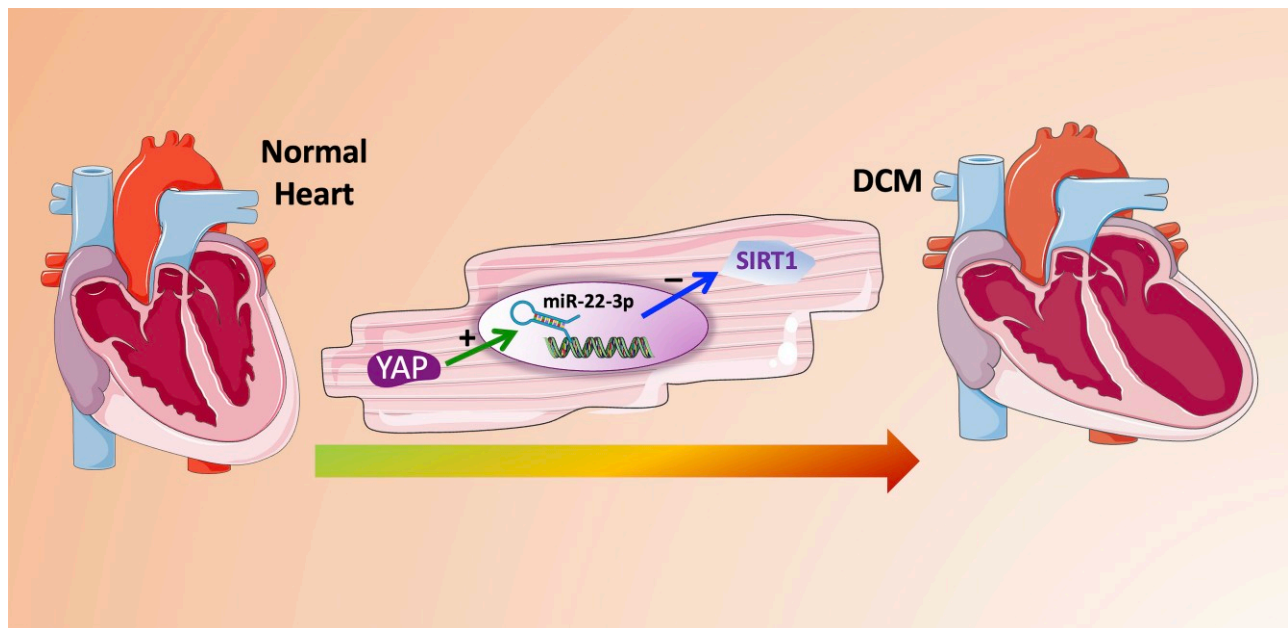


Figure 1—Schematic representation of a key molecular pathway involved in the pathogenesis of DCM.

and extracellular matrix production via CTGF and CYR61 induction (12). The study extends this paradigm into the metabolic domain, positioning YAP as a mechanosensitive and metabolosensitive amplifier of diabetic stress (10).

A major conceptual advance is the identification of SIRT1 as a downstream effector of YAP in DCM (10). The cardioprotective role of SIRT1 is well established: it enhances mitochondrial oxidative capacity, suppresses oxidative stress, and limits fibrotic signaling via deacetylation of PGC-1 α , FOXO, and NF- κ B (13).

YAP inhibition restores SIRT1 expression while SIRT1 knockdown abolishes the protective effects of YAP deficiency (10), providing strong evidence for a linear, functionally relevant pathway that is mechanistically distinct from cancer models where YAP and SIRT1 cooperate to promote survival (14), underscoring the context-dependent wiring of Hippo signaling.

The identification of miR-22-3p as the key mediator linking YAP to SIRT1 repression (Fig. 1) is both mechanistically and biologically plausible: miR-22 has been shown to drive cardiomyocyte hypertrophy and impair metabolic flexibility while directly suppressing SIRT1 in metabolic tissues (15). The demonstration of YAP-dependent induction of miR-22-3p under hyperglycemic stress (10) adds an upstream regulatory layer, consistent with emerging evidence that YAP broadly controls microRNA networks in stress responses (16).

While the study is technically rigorous, several limitations warrant careful consideration. The high-fat diet/streptozotocin model, though widely used, incompletely recapitulates human diabetes; streptozotocin induces partial β -cell toxicity and may exaggerate hyperglycemia-driven stress while underrepresenting early insulin-resistant, hyperinsulinemic states (17). Alternative models (e.g., *db/db* or diet-only insulin

resistance) could clarify whether YAP activation is a general feature of metabolic stress or specific to severe hyperglycemia. The reliance on bulk RNA sequencing obscures cellular heterogeneity (18), as YAP signaling differs profoundly between cardiomyocytes, fibroblasts, and endothelial cells. Single-cell or spatial transcriptomics would be necessary to determine whether the YAP–miR-22-3p–SIRT1 axis is cardiomyocyte autonomous or involves paracrine fibroblast signaling, particularly given the strong profibrotic role of YAP in nonmyocytes (19). Given that SIRT1 is regulated by NAD⁺ availability and post-translational modifications (20), future work should assess deacetylase activity and downstream target acetylation (e.g., PGC-1 α , FOXO1). Functional cardiomyocyte assays (such as calcium handling, sarcomere shortening, or mitochondrial respiration) were not performed; these experiments would have provided mechanistic depth linking molecular remodeling to contractile dysfunction. Finally, human validation is lacking; demonstration of coordinated YAP activation, miR-22-3p upregulation, and SIRT1 suppression in human diabetic myocardium will be essential before translational conclusions can be drawn.

Mechanistically linking Hippo signaling to sirtuin biology through microRNA control advances the field of cardiometabolism beyond descriptive pathology toward mechanistic coherence. A previously unrecognized YAP–miR-22-3p–SIRT1 signaling axis was shown by Hou and associates to integrate metabolic stress with epigenetic repression in DCM. The work provides a strong conceptual framework for future therapeutic strategies targeting stress-responsive transcriptional and epigenetic networks in the diabetic heart.

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